Samira Hajizadeh

sh4635@columbia.edu www.linkedin.com/in/samira-hajizadeh (646) 272-8184

EDUCATION

Columbia University New York City, NY
Doctor of Philosophy May 2030
Computer Science, Advised by Prof. Vishal Misra
Specializing in agentic systems and reliable AI, with current research on reliable and efficient code generation, LLM jailbreaking methods, long-context data retrieval, and mosaic detection.
Teaching Assistant for AI in Context

Columbia University New York City, NY
Master of Science May 2026
Electrical Engineering, Research Track Specialization (GPA 3.91/4)
Recipient of the Mahsa Amini Fellowship
Advanced Master's Research (AMR) Fellow, *Specializing on inference-time optimization of code-generating LLMs*
Course Assistant for Signals and Systems (Fall 24, Fall 25)
Notable Courses: Causal Inference, Deep Learning for NLP, Generative AI, Audio & Speech Processing, Deep Learning, Machine Learning, Algorithms for Data Science

University of Tehran Tehran, Iran
Bachelor of Science Mar 2024
Electrical Engineering, Specialization in Control Engineering (GPA 3.82/4)
Teaching Assistant for Engineering Probability and Statistics (four semesters) and Electric Machines (two semesters)
Notable Courses: Artificial Intelligence, Neural Networks, Intelligent Systems, Advanced Programming

PUBLICATIONS

Retroactive Chain-of-Thought (RetroCoT): Forensic Reconstruction Prompts as a Safety Diagnostic Across Model Generations
Accepted to AdvML-Frontiers × CoTMA workshop at Conference on Language Modeling (CoLM 2026) Workshop
EffiPair: Improving the Efficiency of LLM-generated Code with Relative Contrastive Feedback
Under review at ICLR
Sense and Sensitivity: Examining the Influence of Semantic Recall on Long Context Code Reasoning
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics

RESEARCH EXPERIENCE

Columbia University NYC, NY
Student Research Fellow April 2026 - June 2026
Developed an LLM-based knowledge-graph pipeline for analyzing declassified diplomatic cables, spanning entity and relation extraction, cross-document resolution, Wikidata linking, graph construction, and hierarchical narrative discovery.
Designed evaluation frameworks for measuring extraction accuracy, graph fidelity, and entity-resolution quality.
Built interactive Streamlit and WebGL interfaces for exploring narratives across corpora.

Columbia University NYC, NY
Advanced Master's Research Fellow Oct 2025 - May 2026
Worked on improving code-generating LLMs via inference-time optimization.
Developed EffiPair, an inference-time framework that uses relative contrastive feedback to improve the efficiency of LLM-generated code without model fine-tuning. Achieved up to 1.5× runtime speedup while preserving correctness and reducing token usage by over 90% compared with prior feedback-based methods.

Columbia University NYC, NY
Researcher Jan 2025 - Oct 2025
Co-authored a research paper introducing SemTrace, a framework to measure semantic recall in LLM code generation
Observed that LLMs struggle with reasoning when key information is located mid-context (lost-in-the-middle effect)

Analyzed the divergence between semantic and lexical recall in LLMs

Columbia University NYC, NY
Researcher Jan 2025 - May 2025
Developed and fine-tuned Text-to-Speech (TTS) diffusion models to synthetically generate and analyze healthy and with murmur heart sounds for diagnostic applications
investigated zero-shot and unconditional diffusion trajectories for heart-sound classification, building generative based classifier experiments to reduce labeled-data needs

University of Tübingen Tübingen, Germany
Research Assistant Sep 2024 - May 2025
Reviewed current applications of Large Language Models as conversational agents for therapeutic purposes
Assessed the effectiveness and safety and ethical issues of fully-automated LLM-based psychotherapy

Center of Convergent Technologies Tehran, Iran
Research Assistant Sep 2021 - Feb 2024
Achieved high accuracy in detecting anomalous Fundus images using Out-of-Distribution detection techniques
Diagnosed blood hypertension from retinal Fundus images
Created feature detection algorithms for blood vessel extraction and optic disc and macula detection

WORK EXPERIENCE

Espa Labs SF, CA
Software Engineering Intern July 2026 - Current
Building benchmarking and evaluation datasets for AI-assisted document drafting, using LLM-as-a-judge methods to assess factual accuracy, faithfulness, and system precision. Applying iterative prompt-optimization frameworks, including GEPA and DSPy, to analyze failure cases and improve prompts, agent workflows, tool orchestration, and the end-user drafting experience.
Prompt Engineering - Agentic Frameworks - LLM Evaluation - GEPA - DSPy - Python

Rocket Lawyer SF, CA
AI/ML Intern Jun 2025 - Aug 2025
Conducted prompt engineering and tool orchestration for legal document analysis using multi-agent systems
Developed agent pipelines to assess LLM output quality, such as faithfulness in RAG settings and overall factual accuracy
Designed and implemented an AI Insights Agent to extract document-level insights
PydanticAI - Prompt Engineering - Agentic Frameworks - LLM as a judge - Python

Ludwig Maximilian University Munich, Germany
Data Analyst May 2023 - Jan 2024
Monitored visitor behavior of the Museum of Tomorrow based on gaze patterns
Generated heat maps showing areas of interest in museum exhibits
Conducted statistical analysis on gaze data to provide recommendations for improving museum layout and design

PROJECTS

Retroactive Chain-of-Thought (RetroCoT) for LLM Safety Evaluation June 2026
Designed a model-agnostic, single-turn red-teaming method that reframes harmful requests as forensic reconstruction tasks using presuppositional framing, a forensic-analyst persona, and backward reasoning.
Evaluated four frontier LLMs on 50 AdvBench behaviors using dual LLM judges, human adjudication, and statistical confidence intervals; increased confirmed attack success from 0% to 58% on GPT-4o and from 4% to 52% on GPT-4o-mini.
Conducted component ablations, cross-generation analysis, adversarial-feedback controls, and agentic tool-injection experiments, revealing pragmatic-framing vulnerabilities and bypassing initially resistant GPT-5-family models at rates of 48–52%.
LLM Safety - Red Teaming - AdvBench - Ablation Studies - LLM-as-a-Judge - Agentic AI

Multi-Agent RAG for Financial Q&A and Stock Market Analysis Fall 2024
Applied depth-pruning and LoRA to various LLM backbone models to fine-tune them for financial question answering
Implemented multi-agent retrieval-augmented generation for improving model accuracy

Python - PyTorch - Naive and Agentic RAG - ChromaDB - Langchain - CrewAI

SimCLR paper code implementation and result reproduction Fall 2024

Implemented the code for SimCLR which is a framework for Contrastive Learning of Visual Representations

Python - Tensorflow - Google Cloud Platform - Self-supervised Learning

LANGUAGES AND SKILLS

Python, C++, R

PyTorch, Tensorflow, Librosa, Scikit-learn, NumPy, Pandas, PydanticAI, CrewAI

LLMs, Text-to-Speech Diffusion Models, Computer Vision, Out-of-Distribution Detection, Agentic Frameworks

University of Tehran IEEE Student Branch Deep Reinforcement Learning Workshop